

FPC-16

Fixed-Point Precision Core

Native fixed-point arithmetic for the OrbitEdge-1 mission computer

Group 14 | EEE4120F HPES 2026 | University of Cape Town

Faster. Smarter. Mission-Ready.

MOTIVATION

The StarCore-1 handles Q8.8 addition and subtraction natively, but has no multiply or divide instruction. FPC-16 addresses this by adding single-cycle native FPMUL and FPDIV hardware, focusing effort exclusively where software cannot keep up.

KEY FEATURES

- ✓ **FPMUL (opcode 1110):** 32-bit intermediate product, bits[23:8] extracted
- ✓ **FPDIV (opcode 1111):** Pre-shifted dividend, divide-by-zero protected
- ✓ **Signed Arithmetic:** Two's complement FPMUL & FPDIV support
- ✓ **Memory-Mapped I/O:** 0x00FE input / 0x00FF output via LD/ST
- ✓ **End-to-End Pipeline:** Sensor read → FPMUL/FPDIV → telemetry output
- ✓ **Backward Compatible:** All original StarCore-1 tests pass

PERFORMANCE (Amdahl's Law Analysis)

46-87x

instruction reduction

1 FPMUL vs software
shift-and-add loop

up to 2x

system speedup

at 50% FP workload
(Amdahl's Law)

VERIFICATION RESULTS — 123 / 123 Tests Passing

Testbench	Tests	Covers
ALU_tb	42 / 42	FPMUL, FPDIV, Signed arithmetic
ALU_Control_tb	18 / 18	New opcode decoding
ControlUnit_tb	16 / 16	FPMUL/FPDIV control signals
GPR_tb + InstrMem_tb + DM_tb	42 / 42	Register file & memory modules
StarCore1_tb	5 / 5	Full I/O pipeline verified

TARGET APPLICATION

OrbitEdge-1 CubeSat — 500 km low Earth orbit
Attitude control | Telemetry checksums | Atmospheric sensing
Power budget: < 400 mW | Radiation-tolerant FPGA fabric